A PCM system has **L = 64** quantization levels.

1. Find the **SQNR in dB**
2. Comment on the effect of increasing **L**

First, find the number of bits per sample:

$$n = \log_2 L = \log_2(64) = 6 \text{ bits}$$

For a uniform PCM system, the signal-to-quantization noise ratio is:

$$\text{SQNR}_{dB} = 6.02n + 1.76$$

$$\text{SQNR}_{dB} = 6.02(6) + 1.76$$

$$\boxed{\text{SQNR} \approx 37.88 \text{ dB}}$$

A PCM system uses **256 quantization levels**.

1. Calculate the **number of bits per sample**
2. Determine the **SQNR (in dB)** for a uniform quantizer

1. Number of bits per sample

$$n = \log_2 L = \log_2(256) = \boxed{8 \text{ bits/sample}}$$

---

2. SQNR for a uniform quantizer

For a uniform PCM system, the signal-to-quantization noise ratio is:

$$\text{SQNR}_{\text{dB}} = 6.02n + 1.76$$

Substitute $n = 8$:

$$\text{SQNR}_{\text{dB}} = 6.02(8) + 1.76$$

$$\boxed{\text{SQNR} \approx 49.92 \text{ dB}}$$

A PCM system uses **uniform quantization** with a peak signal amplitude of **5 V** and **128 levels**.

1. Find the **step size**
2. Calculate the **quantization noise power**

1. Step size (Δ)

$$\Delta = \frac{2A}{L}$$

$$\Delta = \frac{2 \times 5}{128} = \frac{10}{128}$$

$$\boxed{\Delta = 0.078125 \text{ V}}$$

2. Quantization noise power

For a uniform quantizer, the mean quantization noise power is:

$$P_q = \frac{\Delta^2}{12}$$

$$P_q = \frac{(0.078125)^2}{12}$$

$$P_q = \frac{0.0061035}{12}$$

$$\boxed{P_q \approx 5.09 \times 10^{-4} \text{ V}^2}$$

A TV signal with a bandwidth of 4.2 MHz is transmitted using binary PCM. The number of quantization levels is 512. Calculate:

1. Code word length
2. Transmission bandwidth
3. Final bit rate
4. Output signal to quantization noise ratio.

Given: $f_m = 4.2$ MHz and $Q = 512$.

1. Code word length (N):

$$Q = 2^N$$

$$\therefore \quad N = \frac{\log 512}{\log 2}$$

$\therefore$ N = 9 bits/word **...Ans.**

2. Transmission bandwidth:

$$B_T = \frac{1}{2} N f_s = \frac{1}{2} N (2 f_m)$$

$$\therefore \quad B_T = 9 \times 4.2 \text{ MHz} = 37.8 \text{ MHz}$$ **...Ans.**

3. Final bit rate (r) :

$$r = N f_s = 9 \times 2 \times f_m = 18 \times 4.2 \text{ MHz}$$

$$\therefore \quad r = 75.6 \text{ Mb/s}$$ **...Ans.**

4. Signal to quantization noise ratio: Since the TV signal is not a sinusoidal signal, let us use the general expression of signal to quantization noise ratio.

$$\left[\frac{S}{N_q}\right] = 4.8 + 6\,N\text{ dB} = 4.8 + (6 \times 9)$$

$$\therefore \quad \frac{S}{N_q} = 58.8\text{ dB} \qquad \textbf{...Ans.}$$

This is the maximum signal to noise ratio that we are expected to get from this system.

A television signal (video and audio) has a bandwidth of 4.5 MHz. This signal is sampled, quantized and binary coded to obtain a PCM signal.

1. Determine the sampling rate if the signal is to be sampled at a rate 20% above the Nyquist rate.
2. If the samples are quantized into 1024 levels, determine the number of binary pulses required to encode each sample.
3. Determine the binary pulse rate (bits per second) of binary coded signal and the minimum bandwidth required to transmit the signal.

Solu.:

Given: $W = 4.5\text{ MHz}$

1. $\text{Sampling rate} = 1.2 \times \text{Nyquist rate} = 1.2 \times 2\,W$

$= 1.2 \times 2 \times 4.5 \times 10^6 = 10.8\text{ MHz}$ **...Ans.**

2. Given that number of quantization levels, $Q = 1024$

But $Q = 2^N$

$\therefore$ Number of binary pulse per word, $N = \log_2 Q$

$\therefore \quad N = \log_2 1024$

$\therefore \quad N = 10$ **...Ans.**

3. Binary pulse rate (bit rate) $= N\,f_s = 10 \times 10.8\text{ MHz} = 108\text{ Mbps}$ **...Ans.**

$\text{Bandwidth} = \frac{1}{2}\text{ bit rate} = 54\text{ MHz}$ **...Ans.**

A compact disc (CD) records audio signals digitally by using PCM. Assume the audio signal bandwidth to be 15 kHz.

1. What is Nyquist rate ?
2. If the Nyquist samples are quantized into L = 65,536 levels and then binary coded,determine the number of binary digits required to encode a sample.
3. Determine the number of binary digits per second (bit/s) required to encode the audio signal.

Solu.:

Given: $W = 15\text{ kHz}$

1. Nyquist rate $= 2\,W = 2 \times 15\text{ kHz} = 30\text{ kHz}$. **...Ans.**
2. Number of quantization levels $Q = 65{,}536$

we know that $Q = 2^N$

$\therefore \quad 2^N = 65{,}536$

$\therefore \quad N = 16$ bits. **...Ans.**

$\therefore$ Number of bits to encode each sample is $N = 16$.

3. Number of bits per second $=$ Number of samples/sec. $\times$ Number of bits/sample

$= 15 \times 10^3 \times 16 = 240$ k bits/sec. **...Ans.**